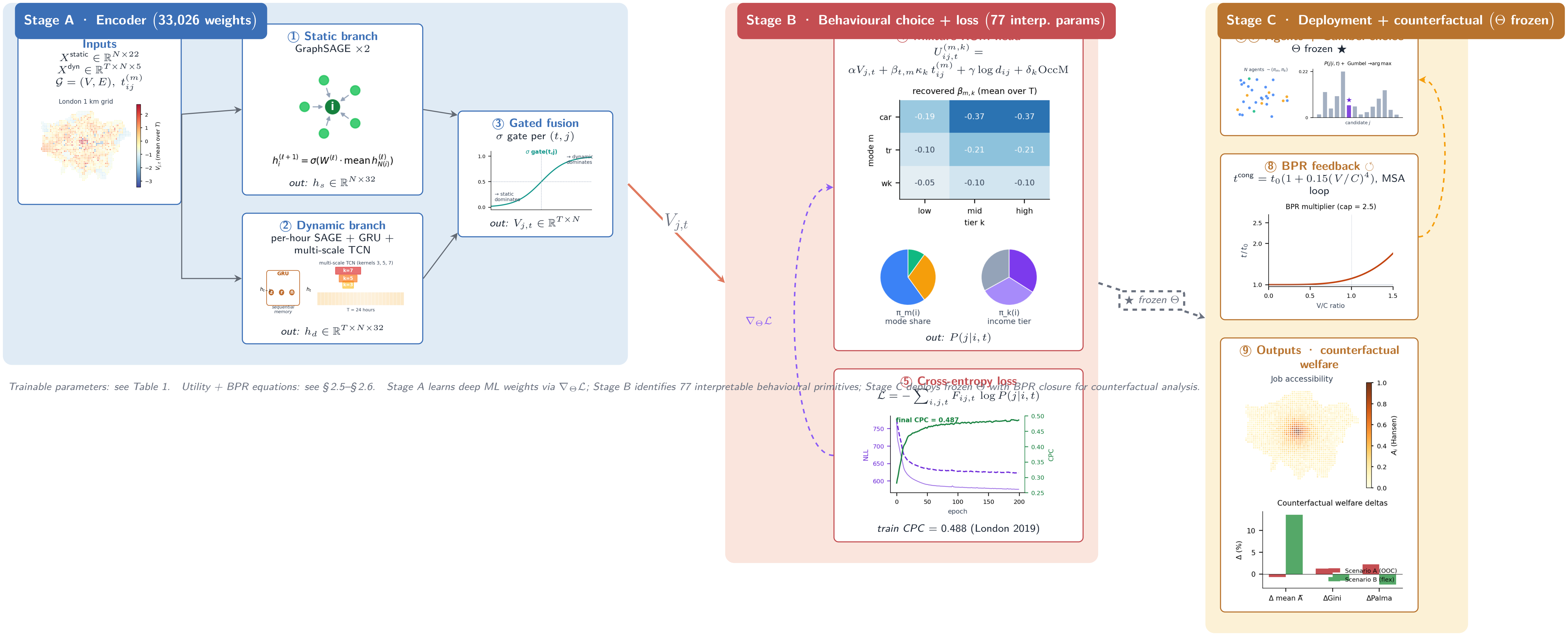


*Trainable parameters: see Table 1. Utility + BPR equations: see § 2.5–§ 2.6. Stage A learns deep ML weights via $\nabla_\Theta \mathcal{L}$; Stage B identifies 77 interpretable behavioural primitives; Stage C deploys frozen Θ with BPR closure for counterfactual analysis.*